

Probability

The setup

Random “experiment”: A well-defined process with an uncertain outcome (for example, toss three fair coins)

Sample space, $\mathcal{S}$: The set of possible outcomes (e.g., $\{ \text{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT} \}$)

Event: A set of outcomes (a subset of $\mathcal{S}$) (e.g., $A = \{\text{exactly one head}\} = \{ \text{HTT, THT, TTH} \}$)

An event is said to have occurred if one of the outcome it contains occurs.

Basic rules of probability

1. $1 \geq \Pr(A) \geq 0$, for any event A .
2. $\Pr(\mathcal{S}) = 1$, for the sample space $\mathcal{S}$.
3. If A and B are *mutually exclusive*, $\Pr(A \text{ or } B) = \Pr(A) + \Pr(B)$.

More rules

$$\Pr(\text{not } A) = 1 - \Pr(A)$$

$$\Pr(A \text{ or } B) = \Pr(A) + \Pr(B) - \Pr(A \text{ and } B)$$

Conditional probability

$$\Pr(A \mid B) = \text{“probability of } A \text{ given } B\text{”} = \Pr(A \text{ and } B) / \Pr(B), \text{ provided } \Pr(B) > 0.$$

Independence

Events A and B are *independent* if $\Pr(A \text{ and } B) = \Pr(A) \Pr(B)$; equivalently, $\Pr(A \mid B) = \Pr(A)$ and $\Pr(B \mid A) = \Pr(B)$.

Still more rules

$$\Pr(A \text{ and } B) = \Pr(B) \Pr(A \mid B) = \Pr(A) \Pr(B \mid A)$$

$$\begin{aligned} \Pr(A) &= \Pr(A \text{ and } B) + \Pr(A \text{ and not } B) \\ &= \Pr(B) \Pr(A \mid B) + \Pr(\text{not } B) \Pr(A \mid \text{not } B) \end{aligned}$$

Bayes rule

$$\begin{aligned} \Pr(A \mid B) &= \Pr(A) \Pr(B \mid A) / \Pr(B) \\ &= \Pr(A) \Pr(B \mid A) / [\Pr(A) \Pr(B \mid A) + \Pr(\text{not } A) \Pr(B \mid \text{not } A)] \end{aligned}$$

Examples

1. Draw two cards (without replacement) from a well-shuffled deck. Calculate the following.
 - (a) $\Pr(\text{2nd card is an ace})$
 - (b) $\Pr(\text{2nd card is an ace} \mid \text{1st card is an ace})$
 - (c) $\Pr(\text{2nd card is an ace} \mid \text{1st card is not an ace})$
 - (d) $\Pr(\text{both cards are aces})$
 - (e) $\Pr(\text{2nd card is an ace} \mid \text{1st card is a heart})$
 - (f) $\Pr(\text{get a pair})$ (i.e., the two cards have the same face).

2. The seeds in Mendel's pea plants were either smooth or wrinkled, the result of a single gene with the smooth allele (S) dominant to the wrinkled allele (a). Similarly, the seeds were either yellow or green, with yellow (Y) dominant to green (y).

The F_1 hybrid seeds produced from crossing two pure-breeding lines (one with smooth, yellow seeds; the other with wrinkled, green seeds) are all smooth and yellow.

Consider growing up an F_1 seed, selfing it, and then taking a random F_2 seed.

(a) $\Pr(F_2 \text{ seed is smooth})$

(b) $\Pr(F_2 \text{ seed is green})$

(c) $\Pr(F_2 \text{ seed is smooth and green})$

(d) $\Pr(F_2 \text{ seed has genotype } SS)$

(e) $\Pr(F_2 \text{ seed has genotype } SS \mid \text{it is smooth})$

3. Consider an urn with 3 red balls and 2 blue balls.

(a) Draw 2 balls *with replacement*.

i. $\Pr(\text{both balls are red})$

ii. $\Pr(\text{the two balls are the same color})$

(b) Repeat the above when the draws are *without replacement*.

i. $\Pr(\text{both balls are red})$

ii. $\Pr(\text{the two balls are the same color})$

4. Roll two fair, six-sided dice (or roll one fair, six-sided dice twice).

(a) **Pr(both dice are 1)**

(b) **Pr(at least one die is a 1)**

5. (A sort of famous 17th century French gambling problem.)

Which of the following is more likely “at least one 1 in 4 rolls of a six-sided die” or “at least one pair of 1’s (snake-eyes) in 24 rolls of a pair of six-sided dice?” Or are they equally likely?

(a) Pr(at least one 1 in 4 rolls of a 6-sided die)

(b) Pr(at least one pair of 1’s in 24 rolls of a pair of 6-sided dice)